

Free Immersions in Critical Dimension from Lie Groups

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September 10, 2026

Abstract

We construct free immersions of $P^3 \times S^1$ and $S^3 \times S^1$ into 14-space. These are the first examples of free immersions in critical dimension of closed manifolds that are not spheres, projective spaces, tori, or surfaces. The idea is to mimic De Leo's construction for m -tori with $m \leq 5$. De Leo constructs a map that is symmetric with respect to a representation of the Lie group formed by the $(m-1)$ -torus; we replace this group with $\mathrm{SO}(3)$. In our examples, the osculating determinant turns out to be constant, unlike in De Leo's.

0 Introduction

0.1 Background

Fix integers $m, q > 0$ and a smooth manifold M of dimension m . Recall that an *immersion* of M into $\mathbf{R}^q$ is a C^1 map $f = (f^1, \dots, f^q): M \rightarrow \mathbf{R}^q$ for which the Jacobian matrix $(\partial_\alpha f^i)_{\alpha,i}$ has maximal rank m everywhere. Following [Gro86, §1.1.4], we say that f is *free* if, more strongly, it is C^2 and the osculating matrix

$$(\partial_\alpha f^i, \partial_{\alpha\beta} f^i)_{(\alpha,i),(\alpha\beta,i)}$$

has maximal rank

$$q_m := m + \binom{m+1}{2} = \frac{m(m+3)}{2}$$

everywhere. Since this forces $q \geq q_m$, we say that q_m is the *critical dimension* for free immersions of m -manifolds.

As defined above, freeness is not an intrinsic property: The rank of the osculating matrix is sensitive to changes of local coordinates on $\mathbf{R}^q$. However, it is preserved by any affine transformation of $\mathbf{R}^q$. More generally, there is an intrinsic notion of freeness for immersions into any smooth manifold equipped with an affine connection.

Beyond [Gro86], one motivation for the study of free immersions is their appearance in Nash's work on embeddings of Riemannian manifolds. See the metric perturbation theorem in [Nas56].

Eliashberg and Gromov showed that if either $q > q_m$ or M is open, then the partial differential relation of being a free immersion of M into $\mathbf{R}^q$ satisfies the

h -principle [GÈ72]. So in this case, such free immersions exist whenever M is parallelizable. This focuses our attention on the existence of free immersions when $q = q_m$ and M is closed and connected.

For the real projective m -space P^m , there is a straightforward construction of a free immersion into $\mathbf{R}^{q_m}$, by showing that the Veronese embedding $P^m \rightarrow P^{q_m}$ factors through an embedded $\mathbf{R}^{q_m}$. Thus, there is also a free immersion of the m -sphere S^m into $\mathbf{R}^{q_m}$ via the quotient map $S^m \rightarrow P^m$.

Perhaps the next simplest example of a closed connected manifold that exists in every dimension m is the m -torus $(S^1)^m$. In the breakthrough work [DL26], De Leo constructs an explicit free immersion

$$F_m: (S^1)^m \rightarrow \mathbf{R}^{q_m} \quad \text{for } m = 2, 3, 4, 5.$$

The key idea in De Leo's work is to choose F_m to be symmetric with respect to a representation of the (real) Lie group $U(1)^{m-1}$ on $\mathbf{R}^q$. More precisely, once we write the torus coordinates as

$$(\vec{x}, z) := (x_1, \dots, x_{m-1}, z) \in (S^1)^{m-1} \times S^1 = (S^1)^m,$$

and identify S^1 with $U(1)$, his F_m takes the form

$$F_m(\vec{x}, z) = R_m(\vec{x}) \cdot v_m(z)$$

for some homomorphism $R_m: U(1)^{m-1} \rightarrow SO(q_m)$ and map $v_m: S^1 \rightarrow \mathbf{R}^{q_m}$. He then reduces the problem to choosing v_m .

From the cases of S^2 , P^2 , and $(S^1)^2$, De Leo obtains a free immersion of any closed surface into $\mathbf{R}^{q_2} = \mathbf{R}^5$ by a gluing argument.

0.2 Results

The purpose of this note is to show that De Leo's method applies beyond tori. In our view, the algebraic structure that makes the homomorphism R_m useful is not just symmetry, but the simple behavior of R_m under differentiation. This behavior, in turn, is an instance of a general fact about how representations of Lie groups transform under invariant vector fields.

The smallest compact Lie groups beyond $U(1)$ are the rotation group $SO(3)$ and its universal cover, the special unitary group $SU(2)$. There is a close analogy between $U(1)$, viewed as the unit circle in the complex plane, and $SU(2)$, viewed as the unit 3-sphere in the 4-space of quaternions. This leads us to hope for analogues of De Leo's free immersions in which $U(1)^{m-1}$ is replaced by $SO(3)$ or $SU(2)$. As these groups are respectively diffeomorphic to P^3 and S^3 , we are proposing to construct free immersions of $P^3 \times S^1$ and $S^3 \times S^1$ into $\mathbf{R}^{q_4} = \mathbf{R}^{14}$. These are our first two results.

Theorem 1. *There is a free immersion $F: P^3 \times S^1 \rightarrow \mathbf{R}^{14}$ of the form*

$$F(g, z) = R(g) \cdot v(z),$$

for some homomorphism $R: \mathrm{SO}(3) \rightarrow \mathrm{GL}_{14}(\mathbf{R})$ and smooth map $v: S^1 \rightarrow \mathbf{R}^{14}$, once we identify P^3 with $\mathrm{SO}(3)$.

In fact, there is a five-parameter family of such maps F . For any fixed choice of parameters, the determinant of the osculating matrix is constant in $(g, z) \in \mathrm{SO}(3) \times S^1$.

Above, the constancy of the osculating determinant seems remarkable to us. In [DL26, §2.1–2.4], the nonconstancy of the determinants implies that further work is needed to check that they are everywhere nonzero.

Corollary 2. *The analogue of Theorem 1 holds with S^3 and $\mathrm{SU}(2)$ in place of P^3 and $\mathrm{SO}(3)$.*

Proof. Via the quotient map $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$. □

Let us mention that the 4-manifolds $P^3 \times S^1$ and $S^3 \times S^1$ are of particular interest in complex geometry. A *primary Hopf surface* is a quotient of $\mathbf{C}^2 - \{0\}$ by an infinite cyclic group generated by some complex-analytic contraction. More generally, a *Hopf surface* is a free quotient of a primary Hopf surface by a finite group. Such complex surfaces form the simplest examples of non-Kähler compact complex manifolds. In [Kod66], Kodaira proved that a complex surface is a primary Hopf surface if and only if it is diffeomorphic to $S^3 \times S^1$. As a corollary, other Hopf surfaces are diffeomorphic to $P^3 \times S^1$.

In light of Theorem 1 and Corollary 2, it is natural to ask whether $P^2 \times S^1$ and $S^2 \times S^1$ also admit free immersions in critical dimension. Recall that a choice of axis lets us identify S^2 with $\mathrm{SO}(2) \backslash \mathrm{SO}(3)$. This suggests that one could bootstrap the S^2 case from Theorem 1 by forming the quotients of appropriate group actions.

0.3 Acknowledgments

I thank Roberto De Leo for explaining his work in the Howard University colloquium. I thank GPT 5.6 Sol for help with simplifying the Ansatz for v in Section 3, as well as other symbolic-algebra calculations. No part of this text was generated by AI.

1 Lie Groups

This section reviews background material on Lie groups that is not strictly needed for [DL26], but needed for our work. It may be found in many places in the literature, as the representation theory of the Lie groups most relevant to us, $\mathrm{U}(1)$ and $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$, is fundamental to the physics of hydrogen-like atoms, though we do not discuss this further. *Henceforth, all vector spaces will be assumed finite-dimensional.*

1.1 Coordinates for Groups

Given any vector space V , we write $\mathrm{GL}(V)$ for its group of linear automorphisms and $\mathfrak{gl}(V)$ for the Lie algebra of $\mathrm{GL}(V)$. If $V = \mathbf{F}^n$, then we write $\mathrm{GL}_n(\mathbf{F})$ and $\mathfrak{gl}_n(\mathbf{F})$ instead. We write $I_n \in \mathrm{GL}_n(\mathbf{F})$ for the identity matrix. We will also use the standard notations $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{SO}(n)$ and $\mathfrak{sl}_n(\mathbf{F})$, $\mathfrak{su}(n)$, $\mathfrak{so}(n)$.

Famously, $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ are related by a degree-2 covering homomorphism $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$. It will be useful to describe the embedding $\mathfrak{su}(2) \subseteq \mathfrak{sl}_2(\mathbf{C})$ and the isomorphism $\mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$ in explicit coordinates. Recall that

$$\begin{aligned}\mathfrak{sl}_n(\mathbf{F}) &= \{\xi \in \mathfrak{gl}_n(\mathbf{F}) \mid \mathrm{tr} \xi = 0\}, \\ \mathfrak{su}(n) &= \{\xi \in \mathfrak{sl}_n(\mathbf{C}) \mid \xi^* + \xi = 0\}, \\ \mathfrak{so}(n) &= \{\xi \in \mathfrak{sl}_n(\mathbf{R}) \mid \xi^* + \xi = 0\}.\end{aligned}$$

Take the standard basis for $\mathfrak{sl}_2(\mathbf{F})$:

$$e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then the following elements form a basis for $\mathfrak{su}(2)$:

$$\begin{aligned}\Xi_1 &:= \begin{pmatrix} 0 & -\frac{\sqrt{-1}}{2} \\ \frac{\sqrt{-1}}{2} & 0 \end{pmatrix} = -\frac{\sqrt{-1}}{2}(e + f), \\ \Xi_2 &:= \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = -\frac{1}{2}(e - f), \\ \Xi_3 &:= \begin{pmatrix} -\frac{\sqrt{-1}}{2} & 0 \\ 0 & \frac{\sqrt{-1}}{2} \end{pmatrix} = -\frac{\sqrt{-1}}{2}h.\end{aligned}$$

(That is, $\Xi_i = -\frac{\sqrt{-1}}{2}\sigma_i$ for $i = 1, 2, 3$, where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices.) Meanwhile, we can define $\mathfrak{so}(3)$ in a coordinate-free way as the Lie algebra freely generated over $\mathbf{R}$ by elements ξ_1, ξ_2, ξ_3 subject to the relations

$$\begin{aligned}[\xi_1, \xi_2] &= \xi_3, \\ [\xi_2, \xi_3] &= \xi_1, \\ [\xi_3, \xi_1] &= \xi_2.\end{aligned}$$

By checking the analogous relations among Ξ_1, Ξ_2, Ξ_3 , we deduce an isomorphism of Lie algebras $\mathfrak{su}(2) \xrightarrow{\sim} \mathfrak{so}(3)$ sending $\Xi_i \mapsto \xi_i$ for $i = 1, 2, 3$.

Since $\mathrm{SU}(2)$ is simply connected, this isomorphism is the differential $d\varpi$ of some homomorphism of Lie groups

$$\varpi: \mathrm{SU}(2) \rightarrow \mathrm{SO}(3).$$

It turns out that ϖ descends to an isomorphism $\mathrm{SU}(2)/\langle \pm I_2 \rangle \xrightarrow{\sim} \mathrm{SO}(3)$.

1.2 Representations

Recall that a *representation* of a Lie group G on a vector space V is an action of G through a homomorphism into $\mathrm{GL}(V)$. An isomorphism of representations is a G -equivariant linear isomorphism. If G is compact, then the representation necessarily decomposes as a direct sum of *irreducible* representations, or *irreps*, up to isomorphism.

Example 3 (The circle). Let $C = \mathbf{R}/2\pi\mathbf{Z}$. For each integer w , there is a complex irrep $\psi_w: C \rightarrow \mathrm{GL}_1(\mathbf{C})$ given by

$$\psi_w(x) := \exp(\sqrt{-1}wx),$$

called the irrep of *winding number* or *weight* w . Note that ψ_1 identifies C with $\mathrm{U}(1)$. Up to isomorphism, these are the only complex irreps.

Example 4 (The torus). Every complex irrep of the d -fold power C^d is a d -fold tensor product of complex irreps of C . So by Example 3, the complex irreps are again 1-dimensional, but now classified by integer d -tuples $\vec{w} \in \mathbf{Z}^d$, still called their *weights*. Explicitly, the complex irrep $\psi_{\vec{w}}: C^d \rightarrow \mathrm{GL}_1(\mathbf{C})$ is given by

$$\psi_{\vec{w}}(\vec{x}) = \exp(\sqrt{-1}\vec{w} \cdot \vec{x}),$$

where $\cdot$ denotes the dot product.

For a complex representation of C^d to be the complexification of a real one, it must decompose into irreps so that the resulting multiset of weights is stable under $\vec{w} \mapsto -\vec{w}$. In particular, $\psi_{\vec{0}}$ complexifies the trivial real irrep, which we will denote by $\varphi_{\vec{0}}$, while for each weight $\vec{w} \neq \vec{0}$, the sum $\psi_{\vec{w}} \oplus \psi_{-\vec{w}}$ complexifies a real irrep $\varphi_{\vec{w}}: C^d \rightarrow \mathrm{GL}_2(\mathbf{R})$. Explicitly,

$$\varphi_{\vec{w}}(\vec{x}) := \begin{pmatrix} \cos \vec{w} \cdot \vec{x} & -\sin \vec{w} \cdot \vec{x} \\ \sin \vec{w} \cdot \vec{x} & \cos \vec{w} \cdot \vec{x} \end{pmatrix}.$$

Note that $\varphi_{\vec{w}}$ and $\varphi_{-\vec{w}}$ are conjugate, hence isomorphic, for any $\vec{w}$. Up to isomorphism, the $\varphi_{\vec{w}}$ for $\vec{w} \in \mathbf{Z}^d / \langle \pm 1 \rangle$ are the only real irreps. Finally, note that in the $d = 1$ case, φ_1 identifies C with $\mathrm{SO}(2)$.

Example 5 ($\mathrm{SU}(2)$). For each integer $k \geq 0$, there is a complex irrep

$$\sigma_k: \mathrm{SU}(2) \rightarrow \mathrm{GL}(W_k),$$

where W_k is the $(k+1)$ -dimensional vector space of homogeneous polynomial functions on $\mathbf{C}^2$ of degree k . Here, $\mathrm{SU}(2)$ acts on W_k by pullback of functions along its action on $\mathbf{C}^2$. We say that σ_k has *weight* k . Up to isomorphism, these are the only complex irreps.

For σ_k to be the complexification of a real irrep, k must be even. In this case, $-I_2 \in \mathrm{SU}(2)$ acts trivially, so σ_k factors through the quotient map $\varpi: \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ from Section 1.1. Up to isomorphism, the real irreps of $\mathrm{SU}(2)$ arising in this way are the only real irreps.

Example 6 ($\mathrm{SO}(3)$). Every complex irrep of $\mathrm{SO}(3)$ arises from some complex irrep of $\mathrm{SU}(2)$ of even weight. That is, in the notation of Example 5, every complex irrep of $\mathrm{SO}(3)$ takes the form $\bar{\sigma}_k: \mathrm{SO}(3) \rightarrow \mathrm{GL}(W_k)$ for some even $k \geq 0$, in which case it is characterized by $\sigma_k = \bar{\sigma}_k \circ \varpi$.

Write $k = 2\ell$. Then σ_k complexifies an irrep of $\mathrm{SU}(2)$ on some $(2\ell + 1)$ -dimensional real vector space V_ℓ . Therefore, $\bar{\sigma}_k$ complexifies an irrep of $\mathrm{SO}(3)$ of the form

$$\rho_\ell: \mathrm{SO}(3) \rightarrow \mathrm{GL}(V_\ell).$$

Explicitly, one may take V_ℓ to be the vector space of homogeneous, degree- ℓ harmonic polynomial functions on $\mathbf{R}^3$. (The restrictions of these functions to S^2 are known as *spherical harmonics*.) Up to isomorphism, the ρ_ℓ are the only real irreps of $\mathrm{SO}(3)$.

1.3 Directional Derivatives

Recall that for a general Lie group G with Lie algebra $\mathfrak{g}$, every element $\xi \in \mathfrak{g}$ defines left- and right-invariant vector fields on G by translation. At $g \in G$, the left-invariant vector field is given by

$$g \cdot \xi = \left. \frac{d}{dt}(g \cdot \exp(t\xi)) \right|_{t \rightarrow 0},$$

where $\exp: \mathfrak{g} \rightarrow G$ is the usual exponential map. More generally, if G acts smoothly from the right on a smooth manifold M , then ξ defines a vector field on M , whose value at $p \in M$ is given by

$$p \cdot \xi := \left. \frac{d}{dt}(p \cdot \exp(t\xi)) \right|_{t \rightarrow 0}.$$

In particular, for any smooth function $f: M \rightarrow N$ with differential $df: TM \rightarrow TN$, we have a directional derivative

$$\partial_\xi f(p) := df(p \cdot \xi) = \left. \frac{d}{dt}f(p \cdot \exp(t\xi)) \right|_{t \rightarrow 0}.$$

For a representation $\rho: G \rightarrow \mathrm{GL}(V)$, this formula becomes

$$(1.1) \quad \partial_\xi \rho(g) = \rho(g) \cdot d\rho(\xi),$$

where $d\rho(\xi) = \left. \frac{d}{dt}\rho(\exp(t\xi)) \right|_{t \rightarrow 0}$.

1.4 Coordinates for Representations

For the generators $\xi_1, \xi_2, \xi_3 \in \mathfrak{so}(3)$ discussed in Section 1.1, and the real irreps $\rho_\ell: \mathrm{SO}(3) \rightarrow \mathrm{GL}(V_\ell)$ discussed in Example 6, we now derive explicit formulas for the differentials $d\rho_\ell(\xi_i) \in \mathfrak{gl}(V_\ell)$ in coordinates on V_ℓ where they become especially clean. Again, these formulas should exist in the literature, but we find it pleasant to provide a more concise account.

By our prior discussion, it suffices to compute the differentials $d\sigma_{2\ell}(\Xi_i) \in \mathfrak{gl}(W_{2\ell})$, then find good coordinates for an $\mathfrak{su}(2)$ -stable real structure on $W_{2\ell}$.

Let x, y be the standard coordinates on $\mathbf{C}^2$, so that $W_{2\ell}$ is the degree- 2ℓ part of $\mathbf{C}[x, y]$. Let $\{w_j\}_{-\ell \leq j \leq \ell}$ be the monomial basis of $W_{2\ell}$ in which

$$w_j := \binom{2\ell}{\ell-j} x^{\ell+j} y^{\ell-j}.$$

For convenience, let $w_{-(\ell+1)}, w_{\ell+1} := 0$.

The $SU(2)$ -action on $W_{2\ell}$ is the restriction of the $GL_2(\mathbf{C})$ -action

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot x^m y^n = (ax + cy)^m (bx + dy)^n.$$

Via the differential of the latter, the elements $e, f, h \in \mathfrak{sl}_2(\mathbf{C})$ act by

$$\left. \begin{aligned} e \cdot w_j &= (\ell + j + 1)w_{j+1}, \\ f \cdot w_j &= (\ell - j + 1)w_{j-1}, \\ h \cdot w_j &= 2jw_j \end{aligned} \right\} \quad \text{for } -\ell \leq j \leq \ell.$$

Expanding the Ξ_i in terms of e, f, h , we obtain

$$\left. \begin{aligned} \Xi_1 \cdot w_j &= -\frac{\sqrt{-1}}{2}((\ell + j + 1)w_{j+1} + (\ell - j + 1)w_{j-1}), \\ \Xi_2 \cdot w_j &= -\frac{1}{2}((\ell + j + 1)w_{j+1} - (\ell - j + 1)w_{j-1}), \\ \Xi_3 \cdot w_j &= -\sqrt{-1}jw_j \end{aligned} \right\} \quad \text{for } -\ell \leq j \leq \ell.$$

In analogy with the formation of cosine and sine from e^{ix} and e^{-ix} , let

$$\left. \begin{aligned} u_j &:= \frac{\sqrt{-1}}{2}(w_j - (-1)^j w_{-j}), \\ v_j &:= \frac{1}{2}(w_j + (-1)^j w_{-j}) \end{aligned} \right\} \quad \text{for } -\ell \leq j \leq \ell.$$

Note that $u_0 = 0$ and $v_0 = w_0$. For convenience, let $u_{\ell+1}, v_{\ell+1} := 0$. Then, in the basis of $W_{2\ell}$ formed by $v_0, u_1, v_1, \dots, u_\ell, v_\ell$, we have

$$\begin{aligned} \Xi_1 \cdot v_0 &= -(\ell + 1)u_1, \\ \Xi_1 \cdot u_j &= \frac{1}{2}((\ell + j + 1)v_{j+1} + (\ell - j + 1)v_{j-1}), \\ \Xi_1 \cdot v_j &= -\frac{1}{2}((\ell + j + 1)u_{j+1} + (\ell - j + 1)u_{j-1}), \\ \Xi_2 \cdot v_0 &= -(\ell + 1)v_1, \\ \Xi_2 \cdot u_j &= -\frac{1}{2}((\ell + j + 1)u_{j+1} - (\ell - j + 1)u_{j-1}), \\ \Xi_2 \cdot v_j &= -\frac{1}{2}((\ell + j + 1)v_{j+1} - (\ell - j + 1)v_{j-1}), \\ \Xi_3 \cdot v_0 &= 0, \\ \Xi_3 \cdot u_j &= jv_j, \\ \Xi_3 \cdot v_j &= -ju_j. \end{aligned}$$

So the real span of this basis forms an $\mathfrak{su}(2)$ -stable real subspace $V_\ell \subseteq W_{2\ell}$. Moreover, the formulas above completely describe the matrices for $d\rho_\ell(\xi_i) = d\sigma_{2\ell}(\Xi_i) \in \mathfrak{gl}(V_\ell)$ with respect to the ordered basis.

Example 7. We list $d\rho_\ell(\xi_1), d\rho_\ell(\xi_2), d\rho_\ell(\xi_3)$, in that order, for small ℓ .

$\ell = 1$:

$$\begin{pmatrix} \cdot & \frac{1}{2} & \cdot \\ -2 & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & \frac{1}{2} \\ \cdot & \cdot & \cdot \\ -2 & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & -1 \\ \cdot & 1 & \cdot \end{pmatrix}.$$

$\ell = 2$:

$$\begin{pmatrix} \cdot & 1 & \cdot & \cdot & \cdot \\ -3 & \cdot & \cdot & \cdot & -\frac{1}{2} \\ \cdot & \cdot & \cdot & \frac{1}{2} & \cdot \\ \cdot & \cdot & -2 & \cdot & \cdot \\ \cdot & 2 & \cdot & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \frac{1}{2} & \cdot \\ -3 & \cdot & \cdot & \cdot & \frac{1}{2} \\ \cdot & -2 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -2 & \cdot & \cdot \end{pmatrix},$$

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & -1 & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -2 \\ \cdot & \cdot & \cdot & 2 & \cdot \end{pmatrix}.$$

$\ell = 3$:

$$\begin{pmatrix} \cdot & \frac{3}{2} & \cdot & \cdot & \cdot & \cdot & \cdot \\ -4 & \cdot & \cdot & \cdot & -1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -\frac{5}{2} & \cdot & \cdot & \cdot & -\frac{1}{2} \\ \cdot & \frac{5}{2} & \cdot & \cdot & \cdot & \frac{1}{2} & \cdot \\ \cdot & \cdot & \cdot & \cdot & -3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 3 & \cdot & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & \frac{3}{2} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ -4 & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & -\frac{5}{2} & \cdot & \cdot & \cdot & \frac{1}{2} & \cdot \\ \cdot & \cdot & -\frac{5}{2} & \cdot & \cdot & \cdot & \frac{1}{2} \\ \cdot & \cdot & \cdot & -3 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -3 & \cdot & \cdot \end{pmatrix},$$

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & -1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -2 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 2 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & -3 \\ \cdot & \cdot & \cdot & \cdot & \cdot & 3 & \cdot \end{pmatrix}.$$

2 Freeness Criteria

Fix a real Lie group G of dimension d , a finite-dimensional real vector space V , a representation $R: G \rightarrow \text{GL}(V)$, and a C^2 map $v: S^1 \rightarrow V$. Let

$$F: G \times S^1 \rightarrow V$$

be defined by

$$(2.1) \quad F(g, z) = R(g) \cdot v(z).$$

We now explain how (2.1) simplifies the study of the freeness of F .

Let $\mathfrak{g}$ be the Lie algebra of G . Choose a basis $(\xi_\alpha)_\alpha$ for $\mathfrak{g}$, and set

$$X_\alpha = dR(\xi_\alpha) \quad \text{for all } \alpha.$$

By (1.1), the first- and second-order partials of R with respect to the left-invariant vector fields induced by the ξ_α are, respectively,

$$\begin{aligned} RX_\alpha & \quad \text{for } 1 \leq \alpha \leq d, \\ RX_\alpha X_\beta & \quad \text{for } 1 \leq \alpha \leq \beta \leq d. \end{aligned}$$

Similarly, the first-order partials of F with respect to these vector fields are

$$\begin{aligned} RX_\alpha \cdot v & \quad \text{for } 1 \leq \alpha \leq d, \\ R \cdot v', \end{aligned}$$

while the second-order partials of F are

$$\begin{aligned} RX_\alpha X_\beta \cdot v & \quad \text{for } 1 \leq \alpha \leq \beta \leq d, \\ RX_\alpha \cdot v' & \quad \text{for } 1 \leq \alpha \leq d, \\ R \cdot v''. \end{aligned}$$

Since the values of R are invertible linear operators, we deduce:

Proposition 8. 1. $G \xrightarrow{R} \text{GL}(V) \rightarrow \mathfrak{gl}(V)$ is a free immersion if and only if the list of matrices

$$(2.2) \quad X_\alpha, \quad X_\alpha X_\beta$$

is linearly independent.

2. F is a free immersion if and only if the list of vectors

$$(2.3) \quad v', \quad v'', \quad X_\alpha \cdot v, \quad X_\alpha \cdot v', \quad X_\alpha X_\beta \cdot v$$

is linearly independent for all $z \in S^1$.

Corollary 9. If F is a free immersion, then $G \xrightarrow{R} \text{GL}(V) \rightarrow \mathfrak{gl}(V)$ is a free immersion.

Proof. Immediate from comparing (2.2) and (2.3). $\square$

Corollary 10. If G is compact, F is a free immersion, and $\dim V = q_{d+1}$, then the G -invariant subspace $V^G \subseteq V$ has dimension at most 2.

Proof. Since G is compact, we can pick an invariant projection $V \rightarrow V^G$. The operators X_α act by zero on V^G , so in the list of vectors (2.3), only v' and v'' can possibly have nonzero projections to V^G . But if $\dim V = q_{d+1}$ and (2.3) is linearly independent, then (2.3) actually forms a basis for V . Then the projections of v' and v'' to V^G must span V^G . $\square$

Example 11. In the notation of Example 4, take $G = C^d$ and $V = \mathbf{R}^q$ and

$$R = \varphi_{\vec{w}^{(1)}} \oplus \cdots \oplus \varphi_{\vec{w}^{(r)}} \oplus \varphi_0^{\oplus s},$$

for some r, s such that $q = 2r + s$ and some weights $\vec{w}^{(i)} = (w_1^{(i)}, \dots, w_d^{(i)}) \in \mathbf{Z}^d$. Taking $\xi_\alpha = \partial_{x_\alpha}$ for $1 \leq \alpha \leq d$, we compute

$$(2.4) \quad X_\alpha = w_\alpha^{(1)} J \oplus \cdots \oplus w_\alpha^{(r)} J \oplus 0_s, \quad \text{where } J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

from which

$$X_\alpha X_\beta = -w_\alpha^{(1)} w_\beta^{(1)} I_2 \oplus \cdots \oplus -w_\alpha^{(r)} w_\beta^{(r)} I_2 \oplus 0_s.$$

(Above, 0_s is the $s \times s$ zero matrix.)

We deduce that for the $X_\alpha X_\beta$ to be linearly independent, it is necessary that whenever a (real) quadratic k -form $Q(\vec{x}) = \sum_{\alpha, \beta} c_{\alpha\beta} x_\alpha x_\beta$ satisfies $Q(\vec{w}^{(1)}) = \cdots = Q(\vec{w}^{(r)}) = 0$, we must have $c_{\alpha\beta} = 0$ for all α, β . In other words, for $C^d \xrightarrow{R} \text{GL}(V) \rightarrow \mathfrak{gl}(V)$ to be free, the evaluation map from the vector space of quadratic d -forms into $\mathbf{R}^r$ that sends

$$Q \mapsto (Q(\vec{w}^{(1)}), \dots, Q(\vec{w}^{(d)}))$$

must be injective. In particular, we must have

$$\binom{d+1}{2} \leq r \leq \lfloor q/2 \rfloor.$$

When $q = q_{d+1}$, this inequality forces $d \leq 4$.

By Corollary 9, the discussion above refines the discussion in [DL26, §2.5], which showed directly that a map $F: (S^1)^m \rightarrow \mathbf{R}^{q_m}$ of the form (2.1), via $C \simeq S^1$, can be free only if $m \leq 5$.

Remark 12. In the setup of Example 11, the freeness of $C^d \xrightarrow{R} \text{GL}(V) \rightarrow \mathfrak{gl}(V)$ also forces the weights $\vec{w}^{(1)}, \dots, \vec{w}^{(r)}$ to be linearly independent as vectors in $\mathbf{R}^d$. For otherwise, they all vanish on some nonzero linear functional λ on $\mathbf{R}^d$. But then the symmetric square of λ is a nonzero quadratic form that vanishes on each weight.

Note that the linear independence of the weights is a tighter constraint than the linear independence of the X_α , due to (2.4) and the bound $\binom{d+1}{2} \leq r$.

Example 13. Take $G = \text{SO}(3)$. In the notation of Example 6, take $V = V_1$ and $R = \rho_1$. Here, one can check that the nine matrices $X_\alpha, X_\alpha X_\beta$ are indeed linearly independent, so ρ_1 gives rise to a free immersion

$$\text{SO}(3) \rightarrow \mathfrak{gl}(V_1) \simeq \mathbf{R}^9.$$

Since $q_3 = 9$, this is a free immersion in critical dimension.

In fact, after identifying $\text{SO}(3)$ with P^3 , one can check that this map is just a rescaled version of the free immersion of P^3 into $\mathbf{R}^9$ that arises from the Veronese embedding of P^3 into P^9 , discussed in the introduction.

3 The $P^3 \times S^1$ Construction

Henceforth, we focus on maps $F: \text{SO}(3) \times S^1 \rightarrow \mathbf{R}^{14}$ of the form (2.1).

3.1 R

The possibilities for R are constrained. In the notation of Example 6, R must be isomorphic to a direct sum of the irreps ρ_ℓ for integers $\ell \geq 0$, where the underlying vector space of ρ_ℓ has dimension $2\ell + 1$. In particular, these dimensions must sum to 14. Moreover, by Corollary 10, the multiplicity of the trivial irrep ρ_0 must be ≤ 2 . So the possibilities are:

decomposition of R into irreps	multiset of irrep dimensions
$\rho_0^2 \oplus \rho_1 \oplus \rho_4$	1, 1, 3, 9
$\rho_0^2 \oplus \rho_2 \oplus \rho_3$	1, 1, 5, 7
$\rho_0 \oplus \rho_1^2 \oplus \rho_3$	1, 3, 3, 7
$\rho_0 \oplus \rho_1 \oplus \rho_2^2$	1, 3, 5, 5
$\rho_1^3 \oplus \rho_2$	3, 3, 3, 5

It is preferable to maximize the multiplicity of ρ_0 , since this makes the determinant needed to check the linear independence in Proposition 8(2) easier to compute. At the same time, it is preferable to keep the irrep dimensions small, to keep the operators X_α and $X_\alpha X_\beta$ in Proposition 8 manageable. We choose

$$(3.1) \quad R = \rho_2 \oplus \rho_3 \oplus \rho_0^2,$$

now putting ρ_0 at the end of the sum for notational consistency with Example 11 and [DL26]. Then the operators X_α take the form

$$X_\alpha = d\rho_2(\xi_\alpha) \oplus d\rho_3(\xi_\alpha) \oplus 0_2,$$

where $d\rho_2(\xi_\alpha)$ and $d\rho_3(\xi_\alpha)$ are written out explicitly in Example 7.

To compute the products $X_\alpha X_\beta$, we simply compute $X_{\alpha\beta}^{(\ell)} := d\rho_\ell(\xi_\alpha) d\rho_\ell(\xi_\beta)$ for $\ell = 2, 3$. In the order

$$X_{11}^{(\ell)}, \quad X_{12}^{(\ell)}, \quad X_{13}^{(\ell)}, \quad X_{22}^{(\ell)}, \quad X_{23}^{(\ell)}, \quad X_{33}^{(\ell)},$$

they are as follows:

$\ell = 2:$

$$\begin{pmatrix} -3 & \cdot & \cdot & \cdot & -\frac{1}{2} \\ \cdot & -4 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & -1 & \cdot \\ -6 & \cdot & \cdot & \cdot & -1 \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & \cdot & \frac{1}{2} & \cdot \\ \cdot & \cdot & -2 & \cdot & \cdot \\ \cdot & -1 & \cdot & \cdot & \cdot \\ 6 & \cdot & \cdot & \cdot & -1 \\ \cdot & \cdot & \cdot & 1 & \cdot \end{pmatrix},$$

$$\begin{pmatrix} \cdot & \cdot & -1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & -1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & -1 \\ \cdot & -2 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -2 & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} -3 & \cdot & \cdot & \cdot & \frac{1}{2} \\ \cdot & -1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & -1 & \cdot \\ 6 & \cdot & \cdot & \cdot & -1 \end{pmatrix},$$

$$\begin{pmatrix} \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -1 \\ \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & 2 & \cdot & \cdot \\ \cdot & -2 & \cdot & \cdot & \cdot \end{pmatrix}, \quad \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & -1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & -1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & -4 & \cdot \\ \cdot & \cdot & \cdot & \cdot & -4 \end{pmatrix}.$$

 $\ell = 3:$

$$\begin{pmatrix} -6 & & & & -\frac{3}{2} \\ & -\frac{17}{2} & & & -\frac{1}{2} \\ & & -\frac{5}{2} & & -\frac{1}{2} \\ & & & -4 & \\ -1 & & & & -4 \\ & -\frac{15}{2} & & & -\frac{3}{2} \\ & & -\frac{15}{2} & & -\frac{3}{2} \end{pmatrix}, \begin{pmatrix} & & -\frac{7}{2} & & \frac{3}{2} \\ & & & & & & -\frac{1}{2} \\ & -\frac{5}{2} & & & \frac{1}{2} \\ 1 & & & & -1 & & \\ & & & 1 & & & \\ & & \frac{15}{2} & & & & -\frac{3}{2} \\ & -\frac{15}{2} & & & \frac{3}{2} & & \end{pmatrix},$$

$$\begin{pmatrix} & & -\frac{3}{2} & & & & \\ & & & -2 & & & \\ & & & & -2 & & \\ -\frac{5}{2} & & & & & -\frac{3}{2} \\ & & -\frac{5}{2} & & & & -\frac{3}{2} \\ & & & -6 & & & \\ & & & & -6 & & \end{pmatrix}, \begin{pmatrix} -6 & & & & \frac{3}{2} & & \\ & -\frac{5}{2} & & & & \frac{1}{2} & \\ & & -\frac{17}{2} & & & & \frac{1}{2} \\ & & & -4 & & & \\ 1 & & & & -4 & & \\ & \frac{15}{2} & & & & & -\frac{3}{2} \\ & & \frac{15}{2} & & & & -\frac{3}{2} \end{pmatrix},$$

$$\begin{pmatrix} & \frac{3}{2} & & & & & \\ & & & -2 & & & \\ & & 2 & & & & \\ & \frac{5}{2} & & & & -\frac{3}{2} \\ -\frac{5}{2} & & & & \frac{3}{2} \\ & & 6 & & \\ & & & -6 & & \end{pmatrix}, \begin{pmatrix} & & & & & & \\ & -1 & & & & & \\ & & -1 & & & & \\ & & & -4 & & & \\ & & & & -4 & & \\ & & & & & -9 & \\ & & & & & & -9 \end{pmatrix}.$$

3.2 v

In De Leo's free immersions of $(S^1)^2$ and $(S^1)^3$ [DL26, §2.1–2.2], the Ansatz for $v: S^1 \rightarrow \mathbf{R}^q$ is simply the parametrization of an ellipse:

$$(3.2) \quad v(z) = A + (\cos z)B + (\sin z)C \quad \text{for some } A, B, C \in \mathbf{R}^q.$$

In his free immersions of $(S^1)^4$ and $(S^1)^5$ [DL26, §2.3–2.4], it takes this form outside of one coordinate.

Moreover, the ellipse above is constructed so that its projection into most irreducible summands of $R: \mathrm{U}(1)^{m-1} \rightarrow \mathbf{R}^q$ remain ellipses.

This suggests that we try an Ansatz of the form (3.2), whose projections onto V_2 and V_3 via (3.1) remain ellipses. That is, we assume that

$$v(z) = \begin{pmatrix} v_2(z) \\ v_3(z) \\ \gamma \cos z \\ \gamma \sin z \end{pmatrix} \quad \text{for some ellipses } v_\ell: S^1 \rightarrow V_\ell \text{ and } \gamma \neq 0.$$

We credit Sol with noticing that our $v(z)$ can have zero entries, in contrast to De Leo's $v(z)$'s: The reason appears to be the severe noncommutativity of the matrices in Example 7 and §3.1. We arrive at the choices

$$v_2 = \begin{pmatrix} a \\ 0 \\ 0 \\ \alpha \cos z \\ \alpha \sin z \end{pmatrix}, \quad v_3 = \begin{pmatrix} b \\ \beta \cos z \\ \beta \sin z \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{for some } a, b, \alpha, \beta \neq 0.$$

Let $c = \cos z$ and $s = \sin z$. Then $\begin{pmatrix} v' & v'' & X_\alpha \cdot v & X_\alpha \cdot v' & X_\alpha X_\beta \cdot v \end{pmatrix}$ is

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & -3a - \frac{\alpha s}{2} & \frac{\alpha c}{2} & \cdot & -\frac{\alpha c}{2} & -\frac{\alpha s}{2} & \cdot \\ \cdot & \cdot & \frac{\alpha c}{2} & -3a + \frac{\alpha s}{2} & \cdot & -\frac{\alpha s}{2} & \frac{\alpha c}{2} & \cdot \\ -\alpha s & -\alpha c & \cdot & \cdot & -2\alpha s & \cdot & \cdot & -2\alpha c \\ \alpha c & -\alpha s & \cdot & \cdot & 2\alpha c & \cdot & \cdot & -2\alpha s \\ \cdot & \cdot & \frac{3\beta c}{2} & \frac{3\beta s}{2} & \cdot & -\frac{3\beta s}{2} & \frac{3\beta c}{2} & \cdot \\ -\beta s & -\beta c & -4b & \cdot & -\beta s & \cdot & \cdot & -\beta c \\ \beta c & -\beta s & \cdot & -4b & \beta c & \cdot & \cdot & -\beta s \\ \cdot & \cdot & -\frac{5\beta s}{2} & -\frac{5\beta c}{2} & \cdot & -\frac{5\beta c}{2} & \frac{5\beta s}{2} & \cdot \\ \cdot & \cdot & \frac{5\beta c}{2} & -\frac{5\beta s}{2} & \cdot & -\frac{5\beta s}{2} & -\frac{5\beta c}{2} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -\gamma s & -\gamma c & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \gamma c & -\gamma s & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

$$\begin{pmatrix} -3a - \frac{\alpha s}{2} & \frac{\alpha c}{2} & \cdot & -3a + \frac{\alpha s}{2} & \cdot & \cdot \\ \cdot & \cdot & -\alpha c & \cdot & -\alpha s & \cdot \\ \cdot & \cdot & -\alpha s & \cdot & \alpha c & \cdot \\ -\alpha c & 6a - \alpha s & \cdot & -\alpha c & \cdot & -4\alpha c \\ -6a - \alpha s & \alpha c & \cdot & 6a - \alpha s & \cdot & -4\alpha s \\ -6b & \cdot & -\frac{3\beta s}{2} & -6b & \frac{3\beta c}{2} & \cdot \\ -\frac{17\beta c}{2} & -\frac{7\beta s}{2} & \cdot & -\frac{5\beta c}{2} & \cdot & -\beta c \\ -\frac{5\beta s}{2} & -\frac{5\beta c}{2} & \cdot & -\frac{17\beta s}{2} & \cdot & -\beta s \\ \cdot & 10b & -\frac{5\beta c}{2} & \cdot & \frac{5\beta s}{2} & \cdot \\ -10b & \cdot & -\frac{5\beta s}{2} & 10b & -\frac{5\beta c}{2} & \cdot \\ -\frac{15\beta c}{2} & \frac{15\beta s}{2} & \cdot & \frac{15\beta c}{2} & \cdot & \cdot \\ -\frac{15\beta s}{2} & -\frac{15\beta c}{2} & \cdot & \frac{15\beta s}{2} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}.$$

We arrive at

$$\begin{aligned} \det \begin{pmatrix} v' & v'' & X_\alpha \cdot v & X_\alpha \cdot v' & X_\alpha X_\beta \cdot v \end{pmatrix} &= -\frac{15^4}{2} ab\alpha^4 \beta^6 \gamma^2 (c^2 + s^2)^6 \\ &= -\frac{50625}{2} ab\alpha^4 \beta^6 \gamma^2. \end{aligned}$$

Since $a, b, \alpha, \beta, \gamma \neq 0$, this is simply a nonzero constant. By Proposition 8, this proves Theorem 1.

Remark 14. The main constraint on the sine and cosine entries in v_2 and v_3 is that they cannot appear in the same indices of the two maps. For instance, if we instead take

$$v_2 = \begin{pmatrix} a \\ 0 \\ 0 \\ \alpha \cos z \\ \alpha \sin z \end{pmatrix}, \quad v_3 = \begin{pmatrix} b \\ 0 \\ 0 \\ \beta \cos z \\ \beta \sin z \\ 0 \\ 0 \end{pmatrix},$$

then $X_3 v_\ell = 2v'_\ell$, from which $X_\alpha v' = 2X_\alpha X_3 v$, violating linear independence.

To explain conceptually why $X_3 v_\ell = 2v'_\ell$: In the notation of §1.4, the choices above put the trigonometric entries as the coefficients of the basis vectors of V_2 and V_3 labeled u_2, v_2 . In the notation of Example 3, these vectors span an irreducible summand of weight 2 in the restrictions of ρ_2 and ρ_3 to C , once we identify C with the 1-parameter subgroup $\text{SO}(2) \subseteq \text{SO}(3)$ generated by ξ_3 . So on these vectors, $X_3 = d\rho_\ell(\xi_3)$ acts by $2J$, where J is defined as in (2.4).

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